Slivka and Slivka-bio

A lightweight, unopinionated framework for executables as web services, applied to bioinformatics

Mateusz Warowny, Stuart MacGowan, Geoff Barton and Jim Procter

University of Dundee

Jalview

The Dundee Resource for Sequence Analysis and Structure Prediction
http://www.compbio.dundee.ac.uk/drsasp.html

UKRI Biotechnology and Biological Sciences Research Council

Why make Executables into Web Services ?
- Access without local installation
- Take advantage of distributed/cloud compute
- Simplify integration with analysis platforms

Why did we create Slivka ?
The Dundee Resource for Sequence Analysis and Structure Prediction provides access to a range of bioinformatics methods, some as web apps, others as programmatic services for the Jalview Multiple Sequence Alignment and Analysis Workbench. Slivka is a new framework that we have developed to provide the next generation of services for the Jalview Desktop Application and JalviewJS, the JavaScript implementation of Jalview.

Why do we need yet another web services framework ?
Jalview's current web services provide access to bioinformatics tools via Java SOAP web services – provided by JABAWS 2.2 and some legacy systems. These are difficult to maintain and extend for larger datasets. SOAP services are also technically complicated, and clients are not easy to implement in JalviewJS, the JavaScript version of Jalview.

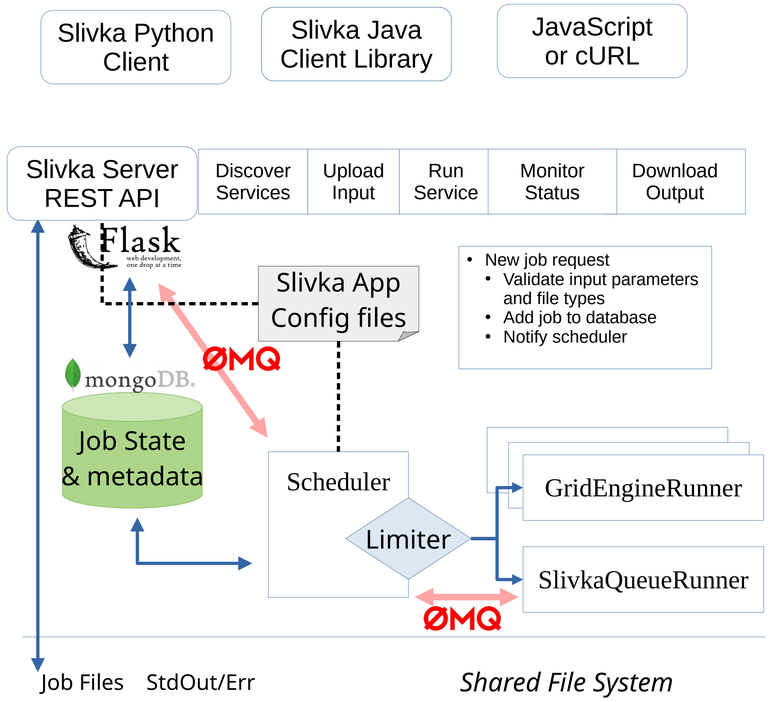
Surely something like Slivka exists already ?
Surprisingly, no. Plenty of systems (e.g. Galaxy) provide APIs for remote command execution, but this is often not their main focus. A small number of 'command line tool as web services' systems have also been developed, including our own. Most of these are unmaintained, however, or lack the flexibility needed for Jalview's services. There are also many excellent web service APIs provided by national and international facilities such as EMBL-EBI, but these are not designed to be easily installed on other systems.

Slivka-bio – a new services stack for Jalview 2.12
Slivka is designed to facilitate easy creation of new types of service, and customisation to support features specific to a particular tool. Jalview's web service architecture has been refactored to take advantage of this approach.

Semantic Service Metadata
Slivka Service's metadata can include ***classifiers***. In slivka-bio, these are drawn from the EDAM ontology. Jalview's client uses these classifiers to determine where the service should be shown in Jalview's Web Services menu, the required **input data** and **expected outputs**.

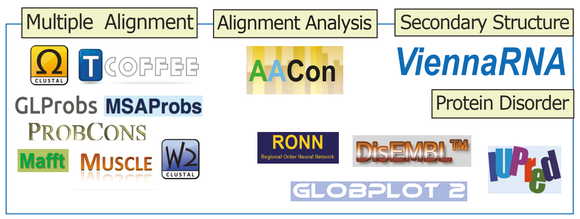
```yaml
slivka-version: 0.8.0
name: IUPred
description: Prediction of Intrinsically Unstructured Proteins
author: Zsuzsanna Dosztányi, Veronika Csizmók, Péter Tompa and István Simon
version: '1.0'

classifiers:
- 'Topic :: Computational biology :: Sequence analysis'
- 'Operation :: Analysis :: Sequence analysis :: Protein sequence analysis'
```

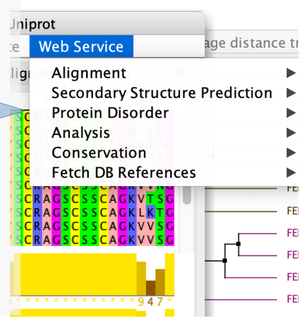


Locally installed command line tools

Tools provided as services by Slivka Bio




```
case "sequence alignment analysis (conservation)":
  newInstance = new SlivkaAnnotationServiceInstance(client,
          service, Category.CATEGORY_CONSERVATION);
  break;
case "protein sequence analysis":
  newInstance = new SlivkaAnnotationServiceInstance(client,
          service, Category.CATEGORY_DISORDER);
  break;
case "protein secondary structure prediction":
  newInstance = new SlivkaAnnotationServiceInstance(client,
          service, "Secondary Structure Prediction");
  break;
case "multiple sequence alignment":
  newInstance = new SlivkaMsaServiceInstance(client, service,
          Category.CATEGORY_ALIGNMENT);
  break;
```



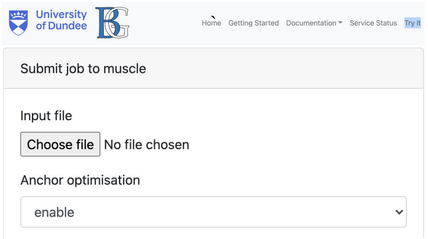
More Information

https://www.compbio.dundee.ac.uk/slivka/

Slivka's homepage provides server status updates, a web interface for executing services (see below), and help documentation on installation and configuring slivka-bio.

Developer and API documentation available from github.

Simple Web form interface
https://www.compbio.dundee.ac.uk/slivka/tryit




```yaml
execution:
  runners:
    uge: !include runners.include.yaml::uge-ram3400M
    local: !include runners.include.yaml::local
  selector: scripts.selectors.TCoffeeSelector
```

```python
class TCoffeeSelector(BaseSelector):
    def setup(self, values):
        self.sequences_data = (
            analyse_sequences_file(
                values['sequence'], 'fasta')
        )

    def limit_local(self, values):
        return (
            self.sequences_data.number <= 15 and
            self.sequences_data.length <= 400
        )

    def limit_uge(self, values):
        return (
            self.sequences_data.number <= 1000 and
            self.sequences_data.length <= 1000
        )
```

Availability

Slivka and Slivka-bio are open source packages released under the Apache 2.0 License.

https://github.com/bartongroup/slivka
https://github.com/bartongroup/slivka-bio

Slivka-bio with Conda

Slivka-bio complete with (nearly) all command-line tools can be installed with the slivka/compbio-services conda environment.

```
conda env create -n my-slivka-bio slivka/compbio-services
conda activate my-slivka-bio
conda install mongodb
mkdir mongo_slivka
mongod -dbpath ./mongo_slivka &
slivka-bio start server &
slivka-bio start scheduler &
Slivka-bio start local-queue &

curl localhost:8000/api/services | less # Check that slivka is up
```

Command-line tools such as IUPred are not yet available in conda. These must be manually installed – see https://bartongroup.github.io/slivka-bio/ for instructions.

Slivka's Runner API

Specified in service definition.

- Submit, monitor & manage jobs
- Inject job-specific resource resource parameters
  - Memory requirements
  - Parallel or GPU resources

Resource limits – work in progress
In Slivka 0.8, runners are selected and configured by evaluating python functions. This allows Cluster or cloud execution parameters to be fine-tuned according to the size of input data, and requested parameters.

We are currently evaluating expression based approaches that will allow runner configuration without creation of python functions.

Inputs and Outputs – data type specifications

Slivka-bio's bioinformatics data types are mapped to flat-file formats supported by Jalview's data import and exporters.

Wrapper scripts are used to generate jalview specific files (such as annotations and features files), but raw outputs are also available.

```yaml
outputs:
  glob:
    path: out.glob
  short:
    path: out.short
  long:
    path: out.long
  jalview-annotations:
    path: iupred.jvannot
    media-type: application/jalview-annotations
  jalview-features:
    path: iupred.jvfeat
    media-type: application/jalview-features
```

```yaml
outputs:
  alignment:
    path: '*.clustalw'
    media-type: application/clustal
  log:
    path: stdout
    media-type: text/plain
```

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